

PRIMETRADE AI SYSTEMS

TECHNICAL & STATISTICAL REPORT

Sentiment-Driven Trading Intelligence & Risk Analytics

REPORT TYPE: Technical Report

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STATUS: COMPLETED & VERIFIED

1. Descriptive Parameter Scope

Central tendency, shape, and variance statistics calculated on closed PnL and market volatility features.

Parameter	Realized Closed PnL Value	Market Volatility Index
Sample Mean	-\$1.68	N/A
Standard Deviation	N/A	N/A
Distribution Skewness	N/A	N/A
Distribution Kurtosis	2.0469	N/A
Minimum Realized Value	-\$15.20	N/A
Median (50th Percentile)	N/A	N/A
Maximum Realized Value	\$8.48	N/A

2. Sentiment Correlation Mappings

Variable Pair Relationship	Pearson Coeff	Pearson p-val	Spearman Coeff	Sig?
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3. Hypothesis Verification Suite

Test Method	Statistic	p-value	Significant?	Target Description
Independent T-Test	N/A	N/A	NO	Fear vs. Greed PnL Means
Mann-Whitney U	N/A	N/A	NO	Fear vs. Greed Rank Comparison
One-Way ANOVA	N/A	N/A	NO	PnL Means across all classes
Chi-Square	N/A	N/A	NO	Sides Mappings vs. Regimes

4. Parameter Confidence Bands (95% Confidence)

Parameter Title	Sample Est	CI Lower	CI Upper	Interval Type
Win Rate	25.00%	4.56%	69.94%	Wilson Score
Average Pnl	-\$1.68	-\$17.37	\$14.01	Student-t
Average Trade Size	\$3.35	-\$6.37	\$13.07	Student-t

5. Statistical Effect Sizes

- * **Cohen's d (Fear vs Greed PnLs):** N/A (Insufficient sample size (requires $n \geq 2$ per group))
- * **Eta-squared (ANOVA Regime Variance):** N/A (ANOVA requires at least 2 groups and total samples $>$ number of groups.)

6. Mathematical Observations

- * **Normality Test:** Shapiro-Wilk test indicates that trade PnL returns follows a normal distribution (p-value: 0.47238).

7. Statistical Charts Appendix

Chart Not Found: pnl_distribution.png

Chart Not Found: sentiment_correlation_scatter.png